MIT ACADEMY OF ENGINEERING

DEPARTMENT OF CSE AIML

Assignment:6

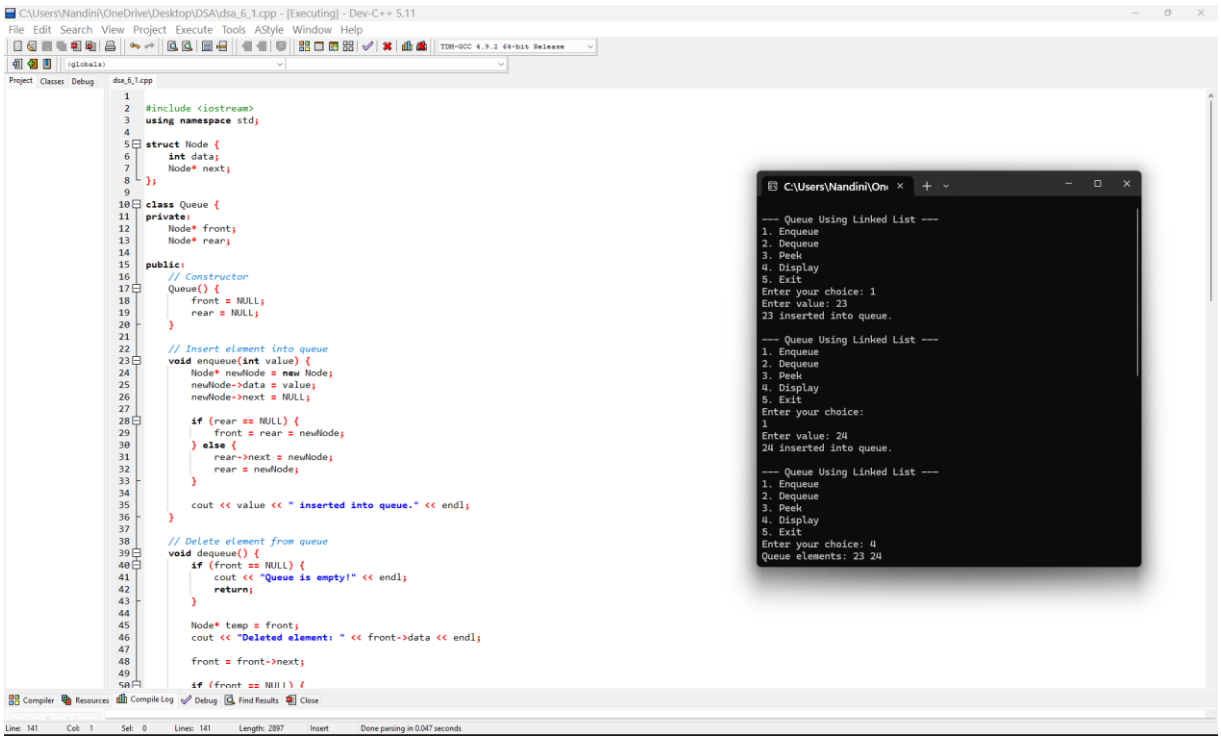
Name : Nandini Ravindra Chimandare

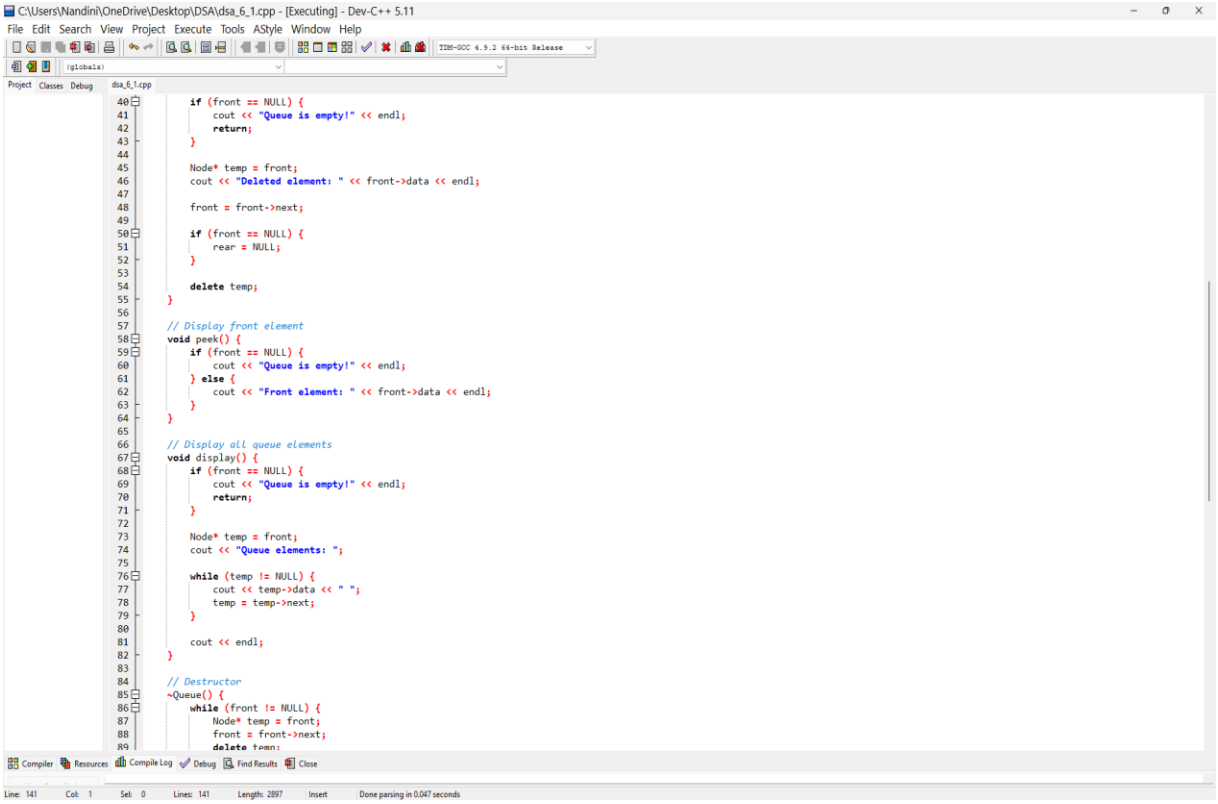
Div/Batch : AIML B1

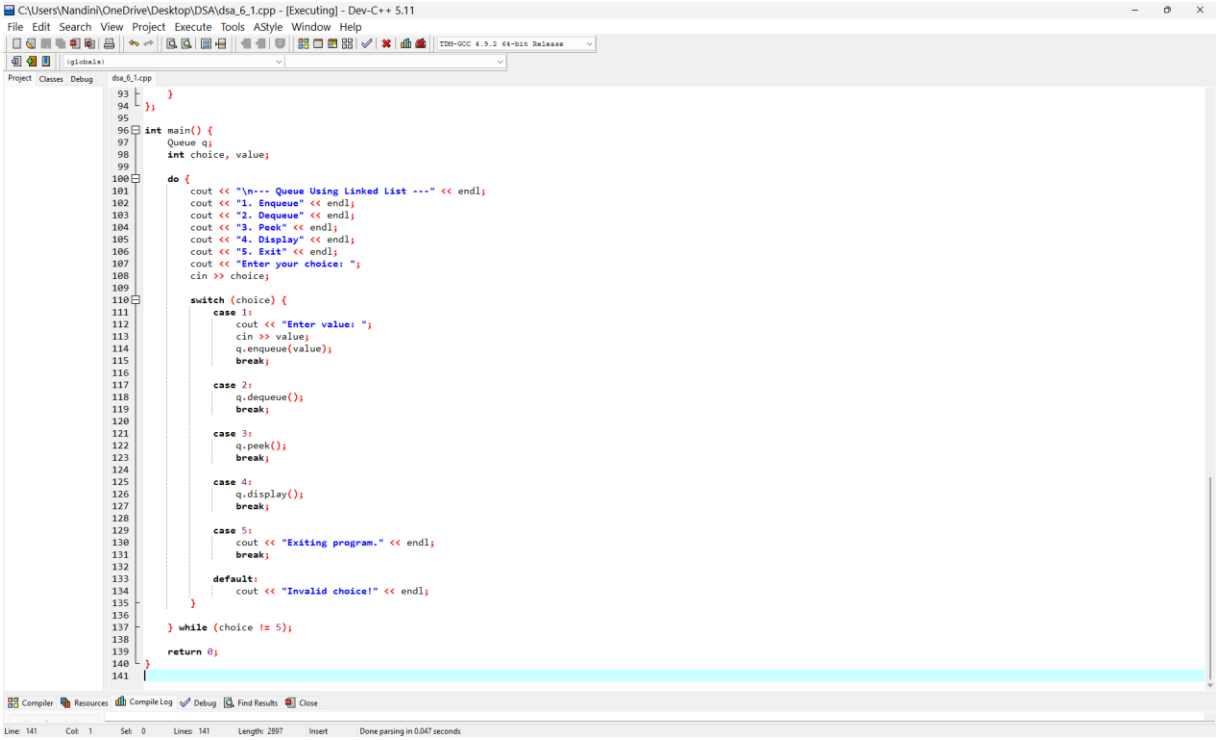
PRN : 202501110108

1.Queue Using Linked List data structure:

- **Implementation:**




C:\Users\Nandini\OneDrive\Desktop\DSA\dsa_6_1.cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 4.9.2 64-bit Release
(globals)
Project Classes Debug
dsa_6_1.cpp
if (front == NULL) {
cout << "Queue is empty!" << endl;
return;
}
Node* temp = front;
cout << "Deleted element: " << front->data << endl;
front = front->next;
if (front == NULL) {
rear = NULL;
}
delete temp;
}
// Display front element
void peek() {
if (front == NULL) {
cout << "Queue is empty!" << endl;
} else {
cout << "Front element: " << front->data << endl;
}
}
// Display all queue elements
void display() {
if (front == NULL) {
cout << "Queue is empty!" << endl;
return;
}
Node* temp = front;
cout << "Queue elements: ";
while (temp != NULL) {
cout << temp->data << " ";
temp = temp->next;
}
cout << endl;
}
// Destructor
~Queue() {
while (front != NULL) {
Node* temp = front;
front = front->next;
Compiler Resources Compile Log Debug Find Results Close
Line: 141 Col: 1 Sel: 0 Lines: 141 Length: 2897 Insert Done parsing in 0.047 seconds


C:\Users\Nandini\OneDrive\Desktop\DSA\dsa_6_1.cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 4.9.2 64-bit Release
(globals)
Project Classes Debug
dsa_6_1.cpp
}
};
int main() {
Queue q;
int choice, value;
do {
cout << "\n--- Queue Using Linked List ---" << endl;
cout << "1. Enqueue" << endl;
cout << "2. Dequeue" << endl;
cout << "3. Peek" << endl;
cout << "4. Display" << endl;
cout << "5. Exit" << endl;
cout << "Enter your choice: ";
cin >> choice;
switch (choice) {
case 1:
cout << "Enter value: ";
cin >> value;
q.enqueue(value);
break;
case 2:
q.dequeue();
break;
case 3:
q.peek();
break;
case 4:
q.display();
break;
case 5:
cout << "Exiting program." << endl;
break;
default:
cout << "Invalid choice!" << endl;
}
} while (choice != 5);
return 0;
}
Compiler Resources Compile Log Debug Find Results Close
Line: 141 Col: 1 Sel: 0 Lines: 141 Length: 2897 Insert Done parsing in 0.047 seconds

Output:


C:\Users\Nandini\On
--- Queue Using Linked List ---
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter value: 23
23 inserted into queue.
--- Queue Using Linked List ---
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice:
1
Enter value: 24
24 inserted into queue.
--- Queue Using Linked List ---
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 4
Queue elements: 23 24
--- Queue Using Linked List ---
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 2
Deleted element: 23
--- Queue Using Linked List ---
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 4
Queue elements: 24
--- Queue Using Linked List ---
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 5
Exiting program.
Process exited after 49.73 seconds with return value 0
Press any key to continue . . .

Code:

```
#include <iostream>
using namespace std;

struct Node {
    int data;
    Node* next;
};

class Queue {
private:
    Node* front;
    Node* rear;

public:
    // Constructor
    Queue() {
        front = NULL;
        rear = NULL;
    }

    // Insert element into queue
    void enqueue(int value) {
        Node* newNode = new Node;
```

```
    newNode->data = value;
    newNode->next = NULL;

    if (rear == NULL) {
        front = rear = newNode;
    } else {
        rear->next = newNode;
        rear = newNode;
    }

    cout << value << " inserted into queue." << endl;
}

// Delete element from queue
void dequeue() {
    if (front == NULL) {
        cout << "Queue is empty!" << endl;
        return;
    }

    Node* temp = front;
    cout << "Deleted element: " << front->data << endl;

    front = front->next;
```

```
    if (front == NULL) {
        rear = NULL;
    }

    delete temp;
}

// Display front element
void peek() {
    if (front == NULL) {
        cout << "Queue is empty!" << endl;
    } else {
        cout << "Front element: " << front->data << endl;
    }
}

// Display all queue elements
void display() {
    if (front == NULL) {
        cout << "Queue is empty!" << endl;
        return;
    }

    Node* temp = front;
    cout << "Queue elements: ";
```

```
        while (temp != NULL) {
            cout << temp->data << " ";
            temp = temp->next;
        }

        cout << endl;
    }

    // Destructor
    ~Queue() {
        while (front != NULL) {
            Node* temp = front;
            front = front->next;
            delete temp;
        }

        rear = NULL;
    }
};

int main() {
    Queue q;
    int choice, value;
```

```
do {
    cout << "\n--- Queue Using Linked List ---" << endl;
    cout << "1. Enqueue" << endl;
    cout << "2. Dequeue" << endl;
    cout << "3. Peek" << endl;
    cout << "4. Display" << endl;
    cout << "5. Exit" << endl;
    cout << "Enter your choice: ";
    cin >> choice;

    switch (choice) {
        case 1:
            cout << "Enter value: ";
            cin >> value;
            q.enqueue(value);
            break;

        case 2:
            q.dequeue();
            break;

        case 3:
            q.peek();
            break;
```

```
            case 4:
                q.display();
                break;

            case 5:
                cout << "Exiting program." << endl;
                break;

            default:
                cout << "Invalid choice!" << endl;
        }

    } while (choice != 5);

    return 0;
}
```